8

Clinical operative procedures: a step-by-step guide

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8.1 Introduction

This chapter illustrates several minimally invasive operative dentistry procedures used for the successful placement of direct plastic restorations in the posterior and anterior dentition. The procedure list cannot be and is not exhaustive, but has been chosen to give the reader the broadest application of the techniques described. The techniques shown are not exclusive—there are many varied operative techniques for removing caries and placing suitable restorations (see Chapters 5 and 6), but the authors feel that the methods described are simple and achievable for most clinical abilities and in most clinical situations. The experienced skilled clinician is able to adapt the myriad of skills outlined in these examples to best fit the developing clinical situation presented to them.

8.1.1 Cavity/restoration classification

There are several classifications of cavities in the dental literature that attempt to correlate site, size of lesion, and disease activity. The oldest, simplest, and probably most universally accepted is Black's classification (see Table 8.1). This classification was originally used to denote the most common sites for caries to develop, so helping to formulate an idea of the individual's caries risk. Nowadays it is used to describe the site of the cavity or restoration, and is useful for descriptive purposes, communicating between dentists, or annotations in dental records. It must be understood that cavities should not be cut with predetermined geometric shapes according to this classification, but the classification should be used according to the final restoration placed, which will be governed by the biological extent of the caries, the type of material used, and other factors (e.g. amount/strength of tooth structure retained, occlusal factors, etc.).

8.1.2 Restoration procedures

The remainder of this chapter will be dedicated to describing and illustrating the practical stages involved in placing certain types of restoration, outlining in detail the clinical procedures involved (see Tables 8.2, 8.3, 8.4, 8.5, 8.6, 8.7, 8.8, 8.9, 8.10, and 8.11). Discussion of the separate steps will be found throughout the preceding chapters, and these links are highlighted throughout. It is assumed that the clinical need for intervention has been appropriately ascertained and that any necessary local analgesia has been administered prior to commencement of the restorative procedures outlined.

Table 8.1 Black's classification of caries lesions—now often misused by dentists as a purely descriptive analysis of the restoration site

Black's class	Site
I	Posterior restoration contained within the occlusal surface (see also Figure 8.1).
II	Posterior restoration including a proximal surface
III	Anterior restoration including a proximal surface only
IV	Anterior incisal edge restoration
V	Buccal cervical surface restoration

Note that with the increased use of adhesive restoratives, many restorations may involve several of the listed sites and therefore need more specific, accurate descriptions.

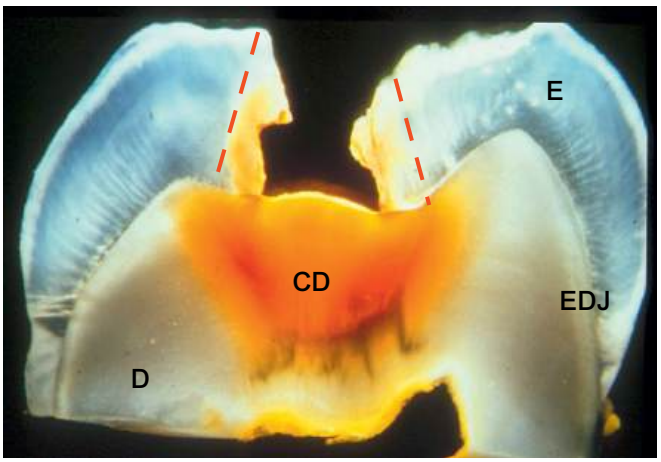


Figure 8.1 A longitudinal section through a cavitated lesion (mICDAS 4), which would also be equivalent to a Black's Class 1 cavity. Note the demineralized, overhanging enamel bordering the cavity, with the undermining spread of the dentine lesion at the EDJ (see Chapter 1 and section 5.9.3 for further discussion of this image). The red dashed lines indicate the extent of enamel removal required in order to achieve a sound enamel margin with supported prism structure, suitable to affect a seal with an adhesive restorative material while still permitting access to the underlying carious dentine. E, enamel; D, dentine; CD, carious dentine; EDJ, enamel–dentine junction.

8.2 Resin-based fissure sealant

Figures 8.2 (a)-(h) illustrate the steps involved in applying fissure sealant. This procedure is also explained in Table 8.2

Table 8.2 Outline of fissure sealant procedure

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/ isolation** (Chapter 5)	**Caries removal/cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Fissure sealant (FS) *Composite*	High caries risk, deep fissure patterns, stagnating plaque Newly erupting molars	Rubber dam (Cotton wool rolls/aspiration)	Fissure debridement with prophy paste and rotating brush Sodium bicarbonate air polishing Bioactive glass air abrasion	Wash and dry using 3-1 syringe (10 s)	*Composite*: 37% orthophosphoric acid-etch enamel fissures (20 s), wash and dry (10 s)	*Composite*: FS flowed into fissure pattern, light cure (470 nm) for 20 s	Remove isolation Check margins with probe Check occlusion with articulating paper
GIC/RM-GIC					*GIC/RM-GIC*: 10% polyacrylic acid conditioning of enamel fissures (15 s), wash and dry (10 s)	*GIC/RM-GIC*: applied into fissure pattern, auto-cured/ light cured (470 nm, 20 s)	



Figure 8.2 (a) Preoperative image of the occlusal surface of LR6 with deep fissures in a high caries risk individual.



Figure 8.2 (b) LR6 cleaned using rotating bristle brush in a slow-speed handpiece and prophy paste.



Figure 8.2 (c) LR6 post-cleaning.



Figure 8.2
(d) 37% orthophosphoric acid etch gel rubbed on to the enamel occlusal surface of LR6 for 20 s.



Figure 8.2
(e) Etch gel washed off for 10 s and air-dried using the 3-1 air/water syringe. Note the frosty appearance of etched occlusal enamel, indicating creation of micro-porosities which aid the micro-mechanical retention of the resin composite fissure sealant.



Figure 8.2
(f) Resin composite fissure sealant applied into fissures using a ball-ended dental probe. The sealant flows easily into the fissure pattern.



Figure 8.2
(g) Resin composite fissure sealant is light cured (470 nm wavelength light for 20 s) to initiate the addition polymerization reaction of the resin monomers.

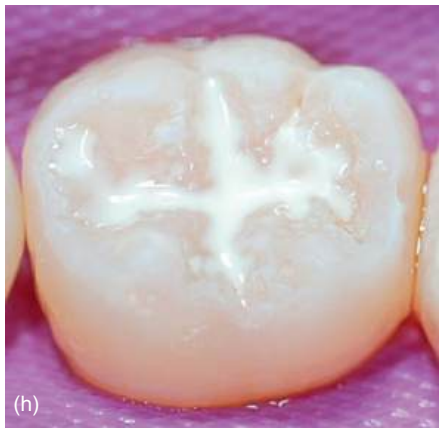


Figure 8.2
(h) Completed occlusal fissure sealant on the LR6.

8.3 Preventive Resin Restoration (PRR); type 3 adhesive (selective enamel etch)

Figures 8.3 (a)-(r) illustrate the steps of a preventative resin restoration (PRR) (type 3 DBA, enamel pre-etch). This procedure is also described in Table 8.3 (see also Chapter 7).

Table 8.3 Outline of preventive resin restoration (PRR)

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/ isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Preventive resin restoration (PRR)	Same as for FS, but evidence of enamel demineralization (mICDAS 1, 2)	Rubber dam (Cotton wool rolls/ aspiration)	Enamel fissure lesion excavated with 330 TC bur, air turbine (Air abrasion—bioactive glass, alumina) Debride remaining fissures with prophy paste/rotating brush	Wash and dry using 3-1 syringe (10 s)	*Resin composite*: 37% orthophosphoric acid-etch enamel fissures (20 s), wash and dry (10 s)	*Composite*: resin flowed into widened fissure, light cure (470 nm) for 20 s	Remove isolation Check margins with probe Check occlusion with articulating paper
					GIC: 10% polyacrylic acid conditioning of enamel fissures (15 s), wash and dry (10 s)	*GIC*: applied into widened fissure, auto-cured/light cured (470 nm. 20 s)	



Figure 8.3 (a) Isolated LR7 with deep stained fissures, cavitated and carious in places in a high caries risk individual (mICDAS 3).



Figure 8.3 (b) Occlusal fissures explored with 330 bur in an air-turbine handpiece.



Figure 8.3 (c) Fissures widened and caries-infected dentine removed distally (soft, sticky). The mesial extent of the fissures opened within enamel.



Figure 8.3
(d) Fissure pattern fully opened distally with caries-affected dentine retained at the cavity floor (scratchy but flaky).



Figure 8.3
(e) Fine diamond bur is used to remove grossly unsupported enamel and lightly bevel the margins.



Figure 8.3
(f) LR7 occlusal enamel pre-etched using 37% orthophosphoric acid etch gel for 15 s.



Figure 8.3
(g) Gel washed off for 10 s and gently air dried (2–3 s).



Figure 8.3
(h) Type 3 DBA applied (2-step, self-etching primer). Self-etching primer rubbed into all cavity surfaces for 10 s.

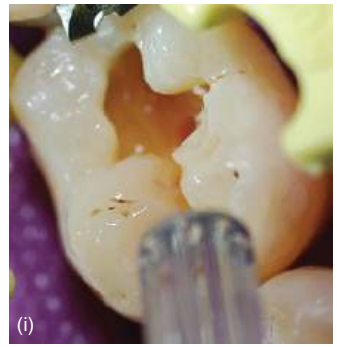


Figure 8.3
(i) Air dried for 5–10 s to evaporate solvent (and thin the layer) until there is no visible 'rippling' of primer on the tooth surface.



Figure 8.3
(j) Type 3 DBA bond/adhesive rubbed on to cavity surfaces.



Figure 8.3
(k) Type 3 DBA bond/adhesive air-thinned as described for Figure 8.5 (i). Note the shiny cavity floor surface which should be seen before light curing (470 nm, 20 s).



Figure 8.3
(l) Flowable resin composite directly introduced into the mesial fissure and light cured.

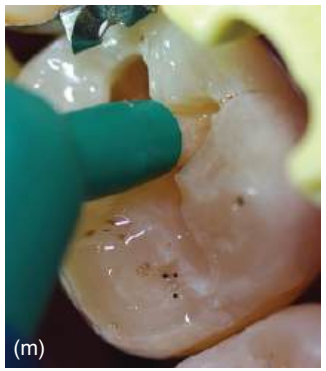


Figure 8.3
(m) Dentine shade resin composite introduced into distal cavity from compule.



Figure 8.3
(o) Enamel shade resin composite directly introduced into the cavity.



Figure 8.3
(q) The occlusion is checked using articulating paper and finished using polishing points and cups (after removing the rubber dam).

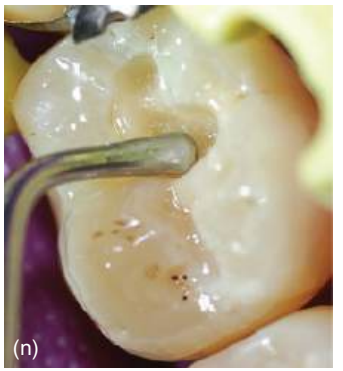


Figure 8.3
(n) Adapted using a burnisher (< 2 mm increment), leaving space for the next enamel shade increment. This is light cured, leaving an uncured air-inhibited layer on the composite surface ready to accept the next increment.

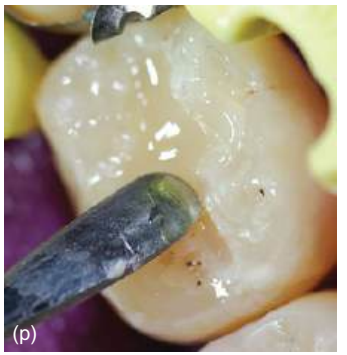


Figure 8.3
(p) Shaped and contoured using a Ward's carver/ pear-shaped burnisher and subsequently light cured.



Figure 8.3
(r) The final finished PRR restoration LR7.

8.4 Posterior occlusal resin composite restoration (Class I); type 3 adhesive

Figures 8.4 (a)-(p) demonstrate a posterior occlusal resin composite restoration (Class I), type 3 adhesive. This procedure is also described in Table 8.4.

Table 8.4 Outline of posterior occlusal adhesive restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Posterior occlusal adhesive restoration: (Class I) All GIC All composite Existing posterior occlusal GIC veneered with composite	High caries risk patient (mICDAS 2); cavitated lesion (mICDAS 3,4) Consider pulp response to sensibility tests Radiographic assessment of pulp proximity	Check occlusion with articulating paper, preoperatively Select tooth-material shade (Vita guides) Place rubber dam	Remove central unsupported/ undermined demineralized enamel with tungsten carbide (TC)/diamond bur, air turbine. Leave sound enamel margin. Peripheral dentine caries excavation (carbon-steel (CS), slow-speed rose-head bur, hand excavator) to affected/sound dentine (lesion depth-dependent). Excavate infected dentine overlying pulp with hand excavator/avoid exposure	Lightly bevel sound enamel margins using fine diamond bur, air turbine Gently round off internal cavity line angles using slow-speed CS rose-head burs Wash and dry cavity (10 s)	*GIC*: 10% polyacrylic acid conditioner rubbed on to cavity walls with micro-brush (15 s), wash and dry (10 s)—removes smear layer and exposes Ca2+ ions for bonding	Mix/dispense *GIC* into cavity, filling from base upwards (to prevent voids/ improve adaptation to cavity walls). Slightly overfill cavity initially. After 60 s pack GIC into the cavity with burnisher/flat plastics, ensuring adequate condensation. Occlusal morphology adapted with flat plastics and excess material removed while GIC still plastic Place *composite* in 2–3 mm increments in stacked angles from base outwards, light curing each increment (20 s). Ensure uncured monomer of the oxygen-inhibited layer on surface of each increment is undisturbed to allow adhesion of next increment. Shape occlusal morphology	*GIC*: wait 3 min for initial set. Check margins/occlusion with articulating paper. Final occlusal adjustments made with diamond burs/stones/sharp scalpel blade. Finish with diamond polishing paste and coat with lightly filled resin for GIC surface protection (dry surface, paint on resin, and light cure (470 nm for 20 s)) *Composites*: Remove rubber dam. Finish surface morphology/ margins with fine diamond burs/discs, check margins/occlusion with articulating paper, polish using diamond grit-impregnated cups/ discs
	Assess occlusal integrity/wear/ marginal failure of > 6-month-old posterior GIC restoration		Remove 2 mm thickness of old, occlusal GIC with diamond/TC bur, air turbine. Leave sound enamel margins		*Composite:* dependent on which DBA (types 1–4) used (see Table 8.12)		

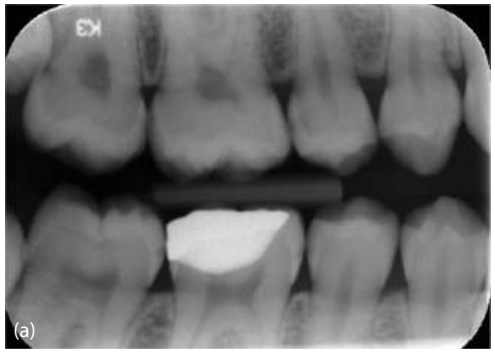


Figure 8.4
(a) Right bitewing radiograph showing multiple early enamel lesions in a high-risk patient.



Figure 8.4
(b) Preoperative view of the occlusal surface of LR7 (mICDAS 2). Rubber dam isolation in place, held anteriorly with medium widget (yellow).



Figure 8.4
(c) Operating view showing rubber dam cut away from nose with paper towel beneath (white). Note that operator's finger rests on lower incisors.

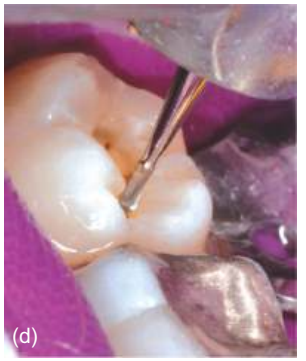


Figure 8.4
(d) 330 bur in air-turbine handpiece used to gain enamel access. Rose-head bur in slow-speed handpiece and excavators used to remove caries-infected dentine.



Figure 8.4
(e) Cavity preparation occlusal LR7, with caries-affected dentine retained over pulp.



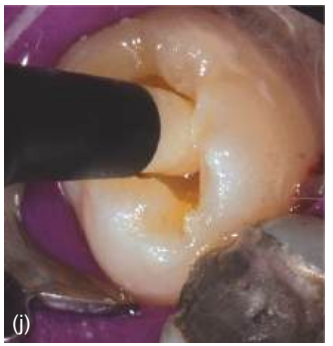


Figure 8.4
(f,g) Adhesive acidic primer rubbed on to cavity surfaces (10 s) using a micro-brush, air-thinned for 5 s and light cured (470 nm, 20 s).





Figure 8.4
(h,i) Adhesive rubbed on to cavity surfaces for 10 s, air-thinned for 5 s, and finally light cured (470 nm, 20 s).



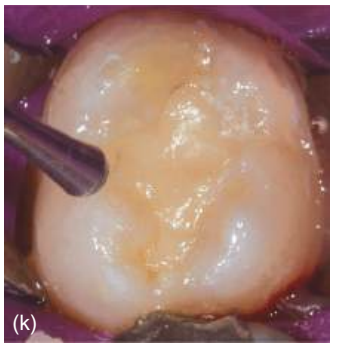


Figure 8.4
(j,k) Resin composite placed directly into cavity and adapted using pear-shaped burnishers.



Figure 8.4
(l) Occlusal composite adjusted using fine diamond rugby-ball-shaped bur.



Figure 8.4
(m) Rubber dam removed and occlusion checked with articulating paper. Note the high spot marked in red on the distal aspect LR7.





Figure 8.4
(n,o) High spot removed/restoration finished using silica-impregnated points/cups. Note how the finishing point has blunted in use. More pressure increases abrasiveness of point but increases wear on the bur.



Figure 8.4
(p) Final resin restoration LR7.

8.5 Posterior proximal resin composite restoration (Class II)

Figures 8.5.1 (a)-(z) and 8.5.2 (a)-(o) illustrate a posterior proximal restoration. This procedure is also described in Table 8.5 (see also Chapter 7).

Table 8.5 Outline of posterior proximal adhesive restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/ isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Posterior proximal adhesive restoration (Class II) Contact point present If approximal lesion directly accessible by instruments (e.g. no adjacent contact, rotated tooth), then treat as buccal cervical restoration (see Table 8.6)	Approximal cavitated lesion/mICDAS 3,4. PA radiograph to ascertain depth; pulp response to sensibility tests	Check occlusion with articulating paper preoperatively Select tooth-material shade (Vita guides) Place rubber dam. Pre-wedge adjacent teeth to aid rubber dam placement. Ensure dam can be placed and held firmly cervical to the base of the carious lesion (clamps, wedgets, floss)	Remove unsupported enamel with TC/ diamond bur (air turbine), accessing proximal lesion through occlusal surface, just medial to the relevant marginal ridge Once access is gained to proximal infected dentine, remove undermined marginal ridge with excavator or bur. Peripheral dentine caries excavation (CS bur, hand excavator) to affected/sound dentine (lesion depth, moisture control-dependent)—'**box' preparation** Excavate infected dentine overlying pulp with hand excavator/ avoid exposure Leave sound enamel periphery, bevelled if possible/avoid gingival papilla trauma	Wash away debris with water from 3-1 syringe and dry (10 s total) Place sectional/ circumferential metal matrix band interproximally ensuring tight adaptation cervically with wedges and contact point formation. Circumferential band/retainers may interfere with rubber dam clamps	*Composite*: dependent upon the DBA used (types 1–4; see Table 8.12) *GIC (if difficulty gaining moisture control)*: 10% polyacrylic acid conditioner rubbed on to cavity walls with micro-brush (15 s), wash and dry (10 s)	Place *resin composite* in 2–3 mm increments in stacked angles from base of box outwards, adapting against walls and matrix band, light curing each increment (470 nm, 20 s). Ensure uncured monomer of the oxygen-inhibited layer on the surface of each cured increment is uncontaminated, allowing adhesion of next increment. Shape occlusal morphology with carving instruments Mix/dispense *GIC* into box base upwards (prevent voids/improve adaptation to cavity walls). Slightly overfill cavity. After 60 s pack GIC into cavity with burnishers/ flat plastics, ensuring adequate condensation. Occlusal morphology adapted with flat plastics	*Composites*: Remove rubber dam. Finish composite surface with fine diamond burs/discs/proximal finishing strips, check margins/ occlusion with articulating paper, polish using diamond grit-impregnated cups/ discs *GIC*: wait 3–4 min for initial set. Check margins/occlusion with articulating paper. Adjustments made with diamond burs/stones/sharp scalpel blade. Finish with diamond polishing paste and coat with lightly filled resin for GIC surface protection

8.5.1 Posterior proximal restoration - type 3 adhesive (selective enamel etch)



Figure 8.5.1
(a) Cavitated LR8 with mesio-occlusal caries (mICDAS 4) and a lesion present in the distal-occlusal pit (mICDAS 2), both requiring operative intervention. Tooth isolated with rubber dam using a steel wingless molar clamp (threaded with dental floss to enable retrieval in case the clamp dislodges from tooth or fractures).



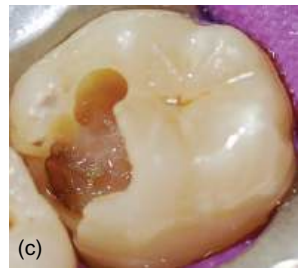


Figure 8.5.1
(b,c) Using a 330 bur in an air-turbine handpiece, access to dentine caries is improved by removing grossly unsupported and undermined enamel at the cavity margins.



Figure 8.5.1
(d) Size 3 rose-head carbon steel bur in a slow-speed handpiece removing peripheral caries at the EDJ.



Figure 8.5.1
(e) Pulpal caries-infected dentine (soft and wet) is carefully removed using a spoon excavator.



Figure 8.5.1
(f) The cavity floor is scratchy and flaky, indicative of caries-affected dentine over the pulp.



Figure 8.5.1
(g) The enamel margin at the base of the mesial box can then be finished using a gingival margin trimmer.

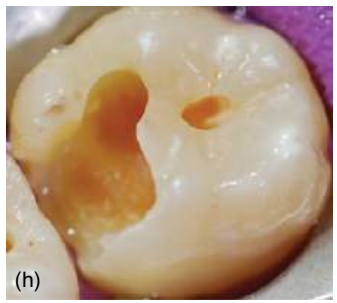


Figure 8.5.1
(h) The enamel margins are lightly bevelled using a multi-fluted tungsten carbide bur. Note how the distal occlusal pit lesion has been accessed in a similar fashion to the preventive resin restoration.



Figure 8.5.1
(i) Placing the sectional metal matrix band in order to reconstitute the mesial wall of the restoration (see Chapter 5). The precurved, contoured band is worked interproximally.





Figure 8.5.1
(j,k) Gingivally contoured plastic wedge is selected and inserted buccally to maintain tight adaptation of band to the mesial surface of LR8.



Figure 8.5.1
(l) Oval retainer with curved tines, placed to secure and adapt band to LR8. This is checked using a sharp dental explorer at the base of the mesial box, ensuring there is no space through which restorative material can extrude.



Figure 8.5.1
(m) Enamel margins pre-etched with 37% orthophosphoric acid etch gel for 15 s, washed off for 10 s, and gently air dried for 2–3 s.



Figure 8.5.1
(n) Self-etching primer (type 3 DBA) is rubbed on to the cavity walls for 10 s.



Figure 8.5.1
(o) Air dried for 5 s to evaporate solvent carrier, until there is no 'rippling' of the primer film on the tooth surface.



Figure 8.5.1
(p) Type 3 bond is applied to cavity walls for 10 s, air-thinned using 3-1 air/water syringe, and light cured for 20 s (470 nm).



Figure 8.5.1
(q) This results in shiny cavity surfaces which must be present prior to placing the first increment of resin composite.



Figure 8.5.1
(r) Occlusal cavity restored with flowable composite. Dentine shade resin composite directly introduced into the mesio-lingual aspect of the cavity LR8.



Figure 8.5.1
(s) Increment is adapted using a pear-shaped burnisher and then light cured for 20 s (470 nm).



Figure 8.5.1
(t) Further 2 mm increments placed and adapted to form the mesial wall of the restoration against firm matrix band and light cured.





Figure 8.5.1
(u,v) Final increments are placed, adapted, and light cured, leaving space for the enamel shade composite to be placed.



Figure 8.5.1
(w) Using Ward's/Half-Hollenback carver, occlusal morphology introduced in the enamel composite, including mesial marginal ridge, fossae, and fissure pattern.







Figure 8.5.1
(x) Retainer removed with clamp, wedge removed buccally, and sectional matrix teased from interproximal area using special tweezers.





Figure 8.5.1
(y) Fine diamond grit/multi-fluted tungsten carbide burs are used to adjust the occlusal/axial surfaces of the restoration after checking the occlusion with articulating paper. Proximal surfaces finished with strips/checked with floss.

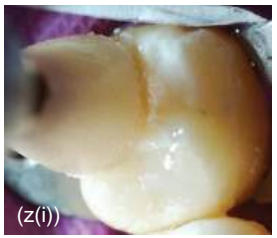
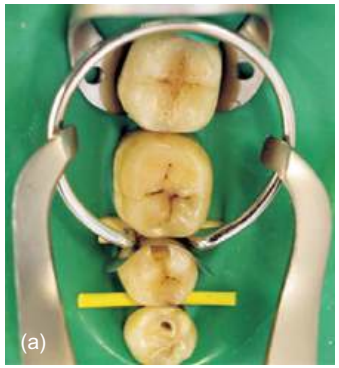






Figure 8.5.1
(z) Composite surface then finished using silica-impregnated cups and points of reducing coarseness. Diamond polishing pastes can help to achieve a high lustre. Final restoration.

8.5.2 Posterior proximal restoration - type 2 adhesive, "moist bonding"



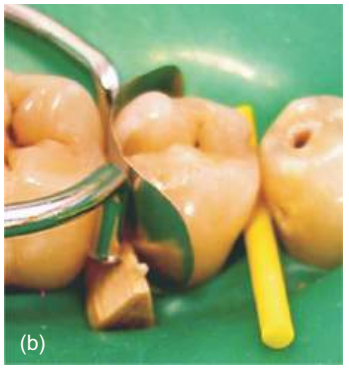


Figure 8.5.2
(a,b) LR5 distal-occlusal cavity (Class II), isolated with rubber dam (wingless clamp LL7 and medium wedget, mesial LL5). Sectional matrix band, wedged gingivally and retained with oval ring.

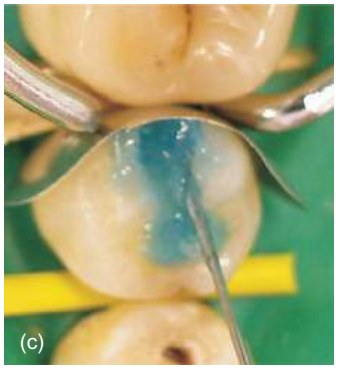


Figure 8.5.2
(c) Cavity acid etched for 20 s.

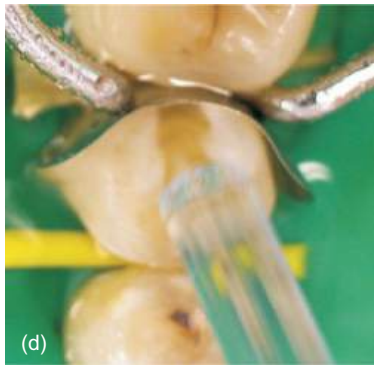


Figure 8.5.2
(d) Cavity washed for 10 s.

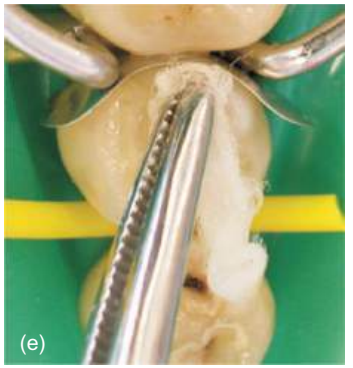


Figure 8.5.2
(e) Cavity blotted dry with cotton wool pledget.

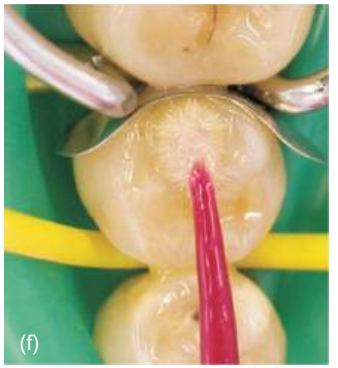


Figure 8.5.2
(f) Type 2 adhesive (primer and bond plus solvent) rubbed on to cavity walls for 5 s.

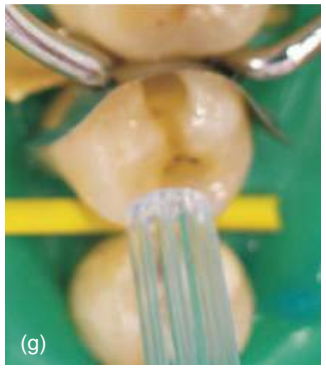


Figure 8.5.2
(g) Type 2 adhesive air-thinned and solvent evaporated for 5 s—look for shiny film on cavity walls.

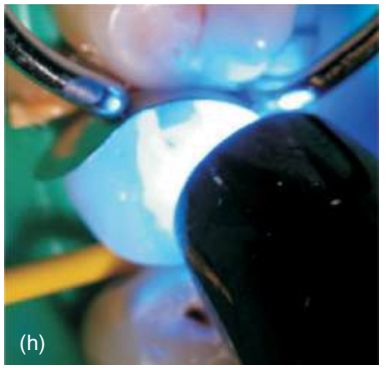
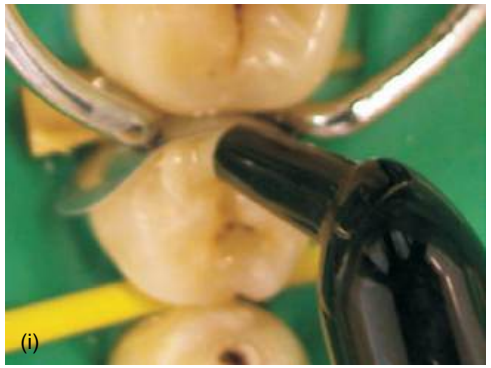


Figure 8.5.2
(h) Type 2 adhesive light cured (470 nm, 20 s).



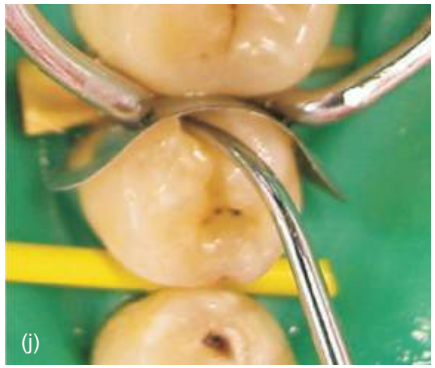


Figure 8.5.2
(i,j) Resin composite placed and adapted into distal box using pear-shaped burnisher. Increments light cured.

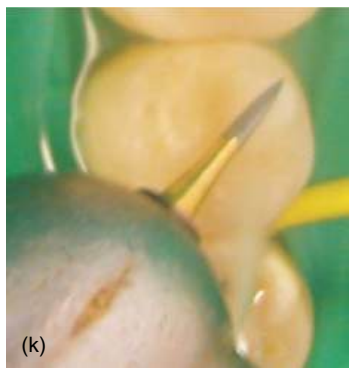






Figure 8.5.2
(k,l,m) Resin composite trimmed and finished using fine diamond finishing burs and abrasive polishing discs.

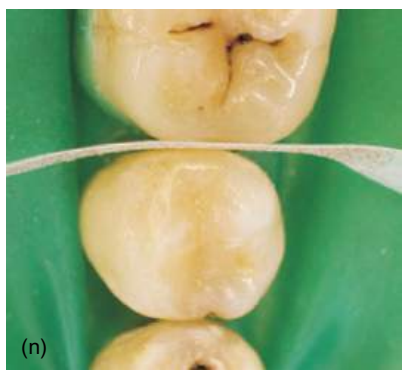




Figure 8.5.2
(n,o) Resin composite finished proximally using abrasive finishing strip, ensuring good distal contour to the final restoration.

8.6 Buccal cervical resin composite restoration (Class V); type 2 adhesive

Figures 8.6 (a)-(n) illustrate a buccal cervical composite restoration. This procedure is also described in Table 8.6.

Table 8.6 Outline of **buccal/lingual** cervical adhesive restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Buccal/ lingual cervical adhesive restoration (Class V)	Discoloured anterior/ posterior tooth wear (causing plaque stagnation or risk of structural weakness) Cavitated carious lesion/ mICDAS 3,4/ pulp response?	Select tooth-material shade Moisture control: rubber dam isolation often difficult as clamp/dam obscures access to lesion margins Simpler cotton wool isolation/aspiration may be necessary	Caries: remove unsupported and/ or undermined demineralized enamel with TC/diamond bur (air turbine) Leave sound enamel margin (may be difficult cervically). Peripheral dentine caries excavation (CS bur, hand excavator) to affected/ sound dentine (lesion depth dependent). Avoid gingival trauma. Excavate infected dentine overlying pulp with hand excavator/avoid exposure Tooth-wear lesion: ideally air abrade exposed lesion surface, removing stain and surface debris	Lightly bevel sound enamel margins using fine diamond bur, air turbine Wash away debris with water from 3-1 syringe and dry (10 s total)	*Composite:* dependent upon the DBA used (types 1–4; see Table 8.12)	Place *resin composite* in 2–3 mm increments in stacked angles from base of cavity outwards, light curing each increment (470 nm, 20 s). Shape occlusal morphology with flat plastic instruments or cervical matrices, removing excess prior to curing	*Composites*: Remove rubber dam. Finish composite surface morphology/margins with fine diamond burs/discs, check margins. and polish using diamond grit-impregnated cups/ discs
					GIC (if difficulty gaining moisture control): 10% polyacrylic acid conditioner rubbed on to cavity walls with micro-brush (15 s), wash and dry (10 s)	Mix/dispense *GIC* filling from base up (prevent voids). Slightly overfill cavity. After 60 s adapt GIC with burnishers/ flat plastics to ensure adequate condensation. Surface may be adapted using cervical matrices	*GIC*: wait 3–4 min for initial set. Final contour adjusted with diamond burs/ stones/sharp scalpel blade. Finish with diamond polishing paste and coat with lightly filled resin for GIC surface protection



Figure 8.6
(a) Buccal cervical cavity in LL3 with plaque stagnation. Gingival margin finishing on dentine.



Figure 8.6
(b) LL3 isolated with rubber dam using a Ferrier clamp to ensure good gingival adaptation of the dam.



Figure 8.6
(c) Acid etch gel placed on all cavity surfaces for 20 s, washed off (10 s) and air dried for 2–3 s ('moist bonding'—no enamel frosting visible). Adhesive then rubbed on to cavity, air-thinned for 5 s, checked for shiny film, and light cured (470 nm, 20 s).



Figure 8.6
(d) Final buccal cervical restoration placed in LL3. Contour achieved using diamond finishing burs and discs.



Figure 8.6
(e) Two small enamel pits on the labial surface UL1 causing an aesthetic concern.

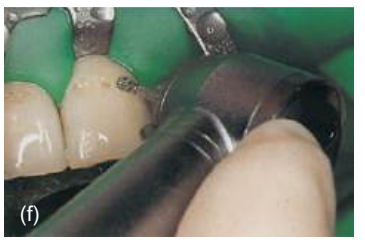


Figure 8.6
(f) Teeth isolated with rubber dam and stain removed with a round diamond bur in the air-turbine handpiece.



Figure 8.6
(g) Acid etch gel placed on the enamel cavities for 20 s, washed off for 10 s, and air-dried for 2–3 s to remove the surface water droplets ('moist bonding').



Figure 8.6
(h) Type 2 adhesive rubbed on to the cavity surfaces, air-thinned to evaporate solvent (5 s), and checked for the presence of a shiny film, then light cured.





Figure 8.6
(i,j) Flowable resin composite (shade checked before rubber dam placement) dispensed into cavities and adapted using a titanium nitride-coated non-stick carver.



Figure 8.6
(k) Composite light cured. Note the orange filter held in front of the light to protect the nurse's and operator's eyes from the intense 470 nm light.

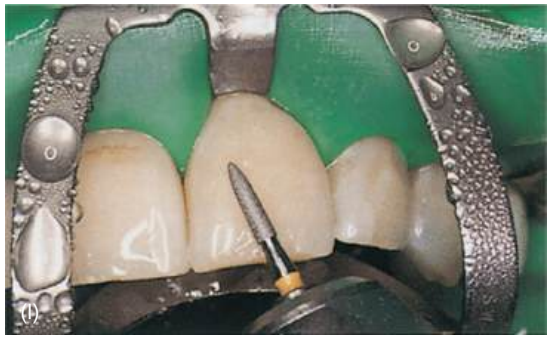


Figure 8.6
(l) Composite adjusted using fine diamond burs with a water spray.



Figure 8.6
(m) UL1 restoration finished using fine abrasive discs.



Figure 8.6
(n) Final restoration UL1.

8.7 Anterior proximal resin composite restoration (Class III); type 2 adhesive

Figures 8.7 (a)-(j) illustrate the steps of a Class III anterior restoration (type 2 DBA). This procedure is also described in Table 8.7.

Table 8.7 Outline of an anterior proximal adhesive restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Anterior proximal adhesive restoration (Class III)	Cavitated carious lesion/mICDAS 3,4/pulp response/ plaque stagnation Replacement of discoloured/failed restoration	Check palatal guiding occlusion with articulating paper preoperatively Select tooth-material shade (Vita guides—dentine/ enamel shades) Place rubber dam. Pre-wedge adjacent teeth to aid rubber dam placement. Ensure dam can be placed and held firmly cervical to the base of the carious lesion (clamps, wedgets, floss)	Caries: remove unsupported/undermined demineralized enamel with TC/diamond bur from palatal aspect (air turbine). Preserve labial enamel wall if possible, to aid aesthetics Leave sound enamel margin (may be difficult cervically). Peripheral dentine caries excavation (CS bur, hand excavator) to affected/ sound dentine (lesion depth dependent). Avoid gingival trauma. Excavate infected dentine overlying pulp with hand excavator/avoid exposure	Lightly bevel sound enamel margins using fine diamond bur, air turbine Wash away debris with water from 3-1 syringe and dry (10 s total) Place 15-mm-long clear matrix strip interproximally and wedge cervically (contact point, cervical adaptation of restorative material)	*Composite*: dependent upon the DBA used (types 1–4; see Table 8.12)	Place *composite* in 2–3 mm increments in stacked angles from base of cavity outwards, light curing each increment (470 nm, 20 s) Consider aesthetic layering techniques with dentine and enamel shades Ensure uncured monomer of the oxygen-inhibited layer on the surface of each cured increment is uncontaminated prior to placement of the next, to allow adhesion of next increment Shape palatal morphology with flat plastic instruments, removing excess prior to curing	Remove wedge, clear matrix band, and rubber dam Finish resin composite surface morphology/ margins with fine diamond burs/ discs/interproximal finishing strips Check margins with probe/protrusive guiding occlusion with articulating paper Polish using diamond grit-impregnated cups/ discs



Figure 8.7
(a) Anterior view of a proximal carious lesion (mICDAS 2) on the distal surface of UL1.



Figure 8.7
(b) The same lesion on UL1; note palatal shadowing.



Figure 8.7
(c) Palatal access cut through enamel using a small round bur in the air-turbine handpiece.



Figure 8.7
(d) The enamel distal marginal ridge cleared through the contact area with the adjacent UL2. Note the brown dentine caries still present.

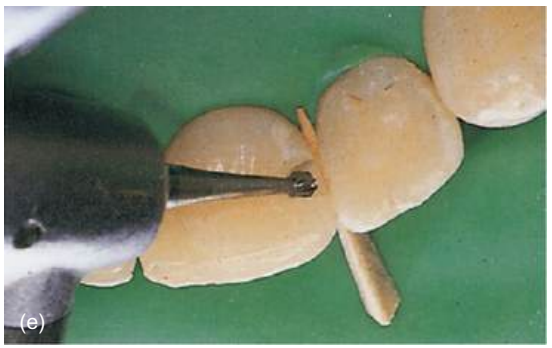


Figure 8.7
(e) All stained dentine excavated leaving yellow hard sound dentine at cavity base for aesthetic reasons, preventing shadowing beneath the final restoration. Labial enamel is preserved.



Figure 8.7
(f) Clear plastic strip is placed interproximally and wedged against the tooth surface. The cavity is etched for 20 s using 37% orthophosphoric acid etch.



Figure 8.7
(g) Etch is rinsed away (10 s) and dried with cotton pledgets ('moist bond technique') before applying the dentine bonding agent and light curing for 20 s.

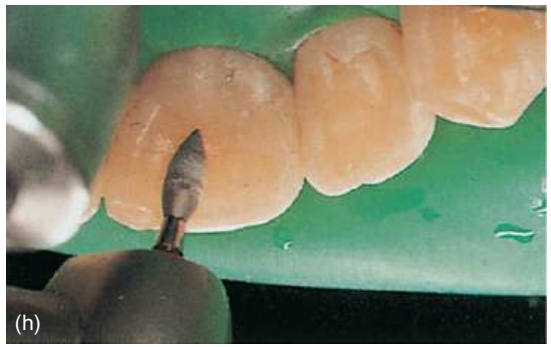


Figure 8.7
(h) Dentine/enamel resin composite placed incrementally and light cured. Palatal surface contour smoothed with rugby-ball-shaped fine diamond composite finishing bur. Occlusion checked with articulating paper



Figure 8.7
(i) Proximal surface finished using abrasive finishing strip (coarse then fine), taking care to preserve the contact point.



Figure 8.7
(j) Final Class III distal restoration UL1.

8.8 Anterior incisal edge/direct labial resin composite veneer (Class IV); type 3 adhesive (selective enamel etch)

Figures 8.8 (a)-(y) illustrate an anterior incisal edge/labial veneer composite (Class IV) - type 3 DBA (enamel pre-etch) procedure. This procedure is also described in Table 8.8.

Table 8.8 Outline of anterior incisal edge/direct veneer restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Anterior incisal edge (Class IV) **Direct veneer restoration**	Replacement of discoloured/failed restoration Traumatic fracture of coronal enamel/dentine Aesthetic masking of intrinsic staining, minor alteration of labial morphology Tooth-wear lesions with superimposed caries—high-risk patients	Check palatal/ protrusive guiding occlusion with articulating paper preoperatively Select tooth-material shade (Vita guides—dentine/enamel shades depending on the composite system) Place rubber dam. Pre-wedge adjacent teeth to aid rubber dam placement. Ensure dam can be placed and held firmly cervical to the base of the carious lesion (clamps, wedgets, floss)	Remove old resin composite restoration with diamond/ TC bur, air turbine or bioactive glass/ alumina air abrasion Excavate soft caries-infected and stained caries-affected dentine with excavators/CS rose-head burs Remove extrinsic staining with sodium bicarbonate air polishing, bioactive glass air abrasion. Achieve sound enamel margins at periphery of cavity	Long, undulated bevel cut into sound enamel on labial surface, fine long-tapered diamond bur, air turbine Short bevel placed on the palatal aspect. Sharp line angles rounded off Wash away debris with water from 3-1 syringe and dry (10 s total) Place 15-mm-long clear matrix strip interproximally and wedge cervically	*Composite*: dependent upon the DBA used (types 1–4; see Table 8.12)	Place composite in 2–3 mm increments in stacked angles from palatal aspect of cavity labially, light curing each (470 nm, 20 s) Can use preformed rigid acrylic/ putty matrix outlining the palatal contour, incisal/mesial/ distal margins; interproximal strips cannot be used with matrices Consider aesthetic layering techniques with dentine and enamel shades Ensure uncured monomer of the oxygen-inhibited layer on the surface of each cured increment is uncontaminated, to allow adhesion of next increment Shape morphology with Ward's/Half-Hollenback carver, removing excess	Remove wedge, clear matrix band, and rubber dam. Finish resin composite surface morphology/ margins with fine diamond burs/ discs/interproximal finishing strips Check margins with probe/protrusive guiding occlusion with articulating paper Polish using diamond grit-impregnated cups/ discs



Figure 8.8
(a) UL1 mesio-incisal enamel–dentine fracture (Class IV) causing sensitivity and an aesthetic concern.

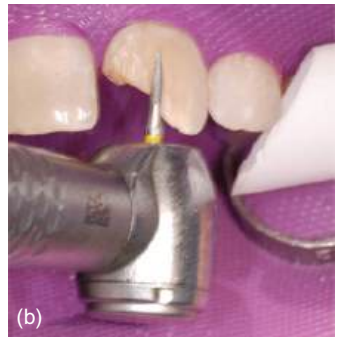


Figure 8.8
(b) Composite shade selected under natural light with hydrated teeth, teeth isolated with rubber dam, and enamel margin prepared with an undulating light bevel on the labial enamel. This is done to prevent a straight-line junction between tooth and final restoration, which would be noticeable clinically.

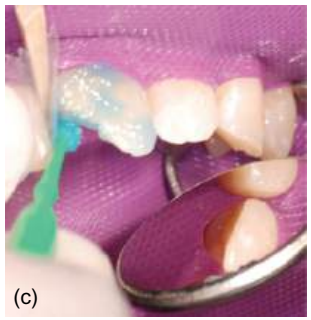


Figure 8.8
(c) Acid-etch gel placed all over enamel margins and labial enamel for 20 s, washed, and dried (10 s).



Figure 8.8
(d) Type 3 DBA primer rubbed over the bonding surfaces for 5 s.

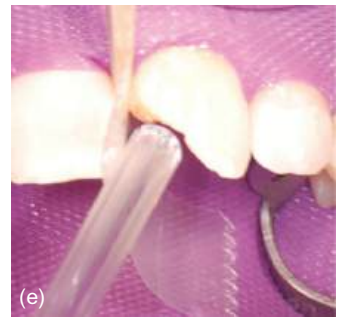


Figure 8.8
(e) Primer air-dried until film no longer ripples (approximately 5 s), removing solvent and air-thinning the film.



Figure 8.8
(f) Type 3 DBA bond rubbed over the cavity surfaces for 5 s, and air-thinned as before and light cured.





Figure 8.8
(g,h) Dentine shade composite placed to form the palatal wall of the final restoration. Note how the clear matrix strip is wedged cervically for marginal adaptation. The underlying mamelon anatomy is mimicked with the composite. This is light cured (470 nm, 20 s).



Figure 8.8
(i) Enamel composite is placed as a roll along the incisal edge of UL1 using a flat plastic.





Figure 8.8
(j,k) Roll of enamel shade composite is then teased cervically across the labial surface with the flat plastic, thinning as it blends with the natural tooth along the undulating bevel.



Figure 8.8
(l) This process permits the placement of a smooth layer of composite labially, which is easily finished later.





Figure 8.8
(m,n) Margins are trimmed using composite finishing fine diamond burs cervically, proximally, and palatally (rugby-ball-shaped bur).





Figure 8.8
(o,p) Incisal occlusion is assessed using articulating paper. Finger placed on UL1 to 'feel' the heavy contact—*fremitus*. A heavy contact is expressed on the mesio-palatal aspect of UL1, with the straight-line trace of the protrusive guidance (towards the incisal edge).



Figure 8.8
(q) This is removed using a fine diamond bur.



Figure 8.8
(r,s,t) The teeth are dried with cotton wool rolls and the occlusion re-checked this time showing a light contact disto-cervically UL1—normal incisal contact.

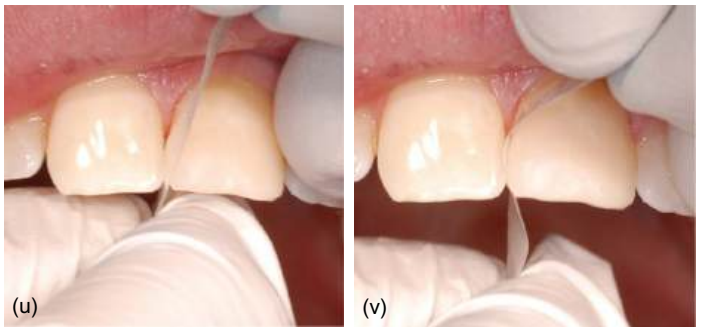


Figure 8.8
(u,v) Proximal coarse/fine finishing strip is worked through the mesial contact, against the restoration subgingivally to remove any excess adhesive, taking care not to remove the contact point.

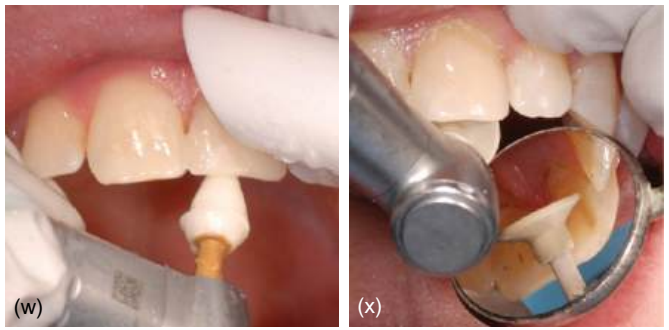


Figure 8.8
(w,x) Composite polished with coarse to fine abrasive points and discs, achieving a final high surface lustre and surface anatomy.



Figure 8.8
(y) Final aesthetic composite restoration UL1 with the patient smiling. Note that the final shade seems slightly darker than the adjacent natural dentition—this is due to dehydration of the natural teeth making them appear whiter. The patient should be warned about this prior to starting the procedure. This will correct itself within a few hours.

8.9 Large posterior bonded amalgam restoration (**courtesy of Dr G Palmer**)

Figures 8.9 (a)-(j) illustrate the steps of restoration of a posterior bonded amalgam. This procedure is also described in Table 8.9 (see also Chapter 6).

Table 8.9 Outline of large posterior amalgam restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapter 5 and 6)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Large posterior amalgam restoration	Heavily broken down coronal tooth structure High load-bearing, large restoration Core build-up for posterior root-filled teeth Difficult moisture control conditions	Check occlusion preoperatively with articulating paper Cotton wool rolls/ aspiration or rubber dam isolation if possible	Cavity/defect already present—access cavity from endodontic treatment, fractured cusps Caries—biologically excavate to sound dentine using hand excavators/CS rose-head burs Remove undermined/ unsupported enamel (diamond/TC burs, air turbine)	Undercut dentine at diametrically opposite aspects of cavity (CS rose-head bur), if remaining tissue permits this Round off sharp internal line angles. Smooth cavity base Indirect pulp capping over blushing pulp with low-viscosity GIC. Wash away debris and dry (10 s) Place circumferential matrix band, wedging the cervical aspect interproximally	*Bonded amalgams*: use auto-/dual-cure resin cement (e.g. Panavia™)	Triturate high-Cu alloy, plug and condense incrementally from the base of the cavity outwards, ensuring adaptation against walls/matrix Slightly overfill cavity and carve back to occlusal level and morphology using carvers (Ward's, Half-Hollenback), flat plastics Work quickly but not hastily!	Remove matrix band. Take care not to fracture/dislodge marginal ridges Check occlusion with articulating paper/ interproximal overhangs with dental floss/finishing abrasive strips Remove high spots with excavator/Mitchell's trimmer or amalgam finishing burs if required Burnish occlusal surface and margins for smooth finish and good adaptation Wipe occlusal surface with slightly damp cotton wool pledget for final surface finish

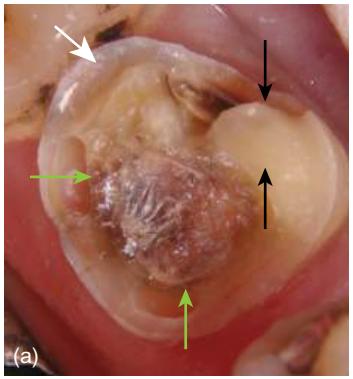


Figure 8.9
(a) Grossly broken down LL6 prepared for an amalgam restoration. Note the two lingual slots (green arrows) and relative undercuts around the mesio-buccal cusp to aid retention (black arrows), and the disto-buccal heavy shoulder preparation to aid distal support (white arrow).



Figure 8.9
(b) Circumferential metal matrix band with retainer placed around tooth and wedged disto-lingually using a wooden wedge to aid adaptation of the distal margin of the final restoration. The band's inner surface is lightly coated with petroleum jelly to facilitate its final removal. To aid retention, a self-etching primer has been rubbed on to the cavity surface and air dried to thin and evaporate the solvent.

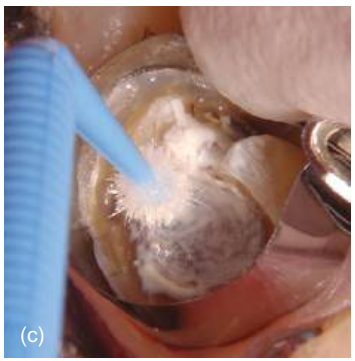


Figure 8.9
(c) A thin chemically cured resin adhesive cement applied using a micro-brush on to the cavity surfaces. This technique is known as a bonded amalgam, assisting retention in very large cavities.

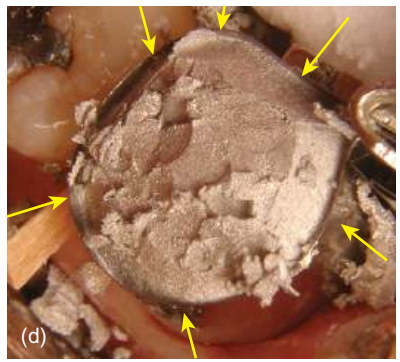


Figure 8.9
(d) The amalgam is condensed into the cavity, filling the volume of the matrix band while keeping in mind the cuspal height of the adjacent teeth. The excess amalgam has been removed around the margins (yellow arrows), thus exposing the edge of the band.

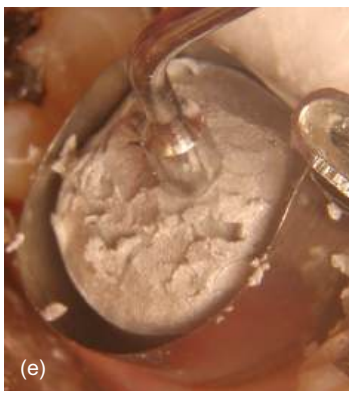


Figure 8.9
(e) After a few minutes, the wedge is removed, and the band is loosened and pushed sideways lingually through the contact area (note the gap, arrowed). The amalgam is supported with a Guy's pattern plugger as the matrix band is eased off.

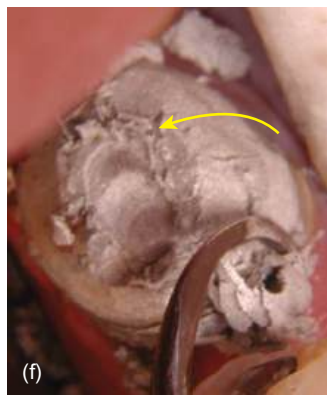


Figure 8.9
(f) Using the curved blade of a sickle scaler, the buccal and lingual walls of the amalgam are contoured, closely following that of the adjacent teeth. The height of the amalgam is reduced to the level of the final cusp tips.

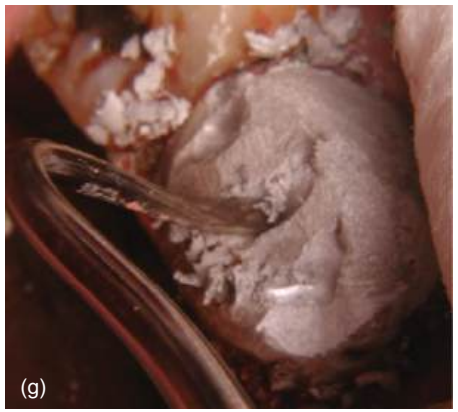


Figure 8.9
(g) The mesial and distal marginal ridges are carved to the height of the adjacent ridges, followed by the marginal ridge and central fossae using a spoon excavator.

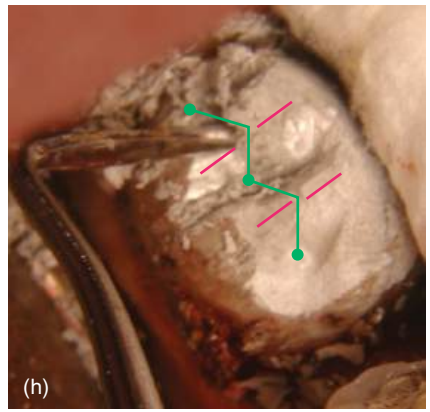


Figure 8.9
(h) Using a Half-Hollenback or Ward's carver, carve in the central fissure pattern (blue line) by joining up the three fossae (green dots). To maintain cusp bulk, try to carve from the fossae up to the transverse ridge (purple lines) and then back down to the adjacent fossa.

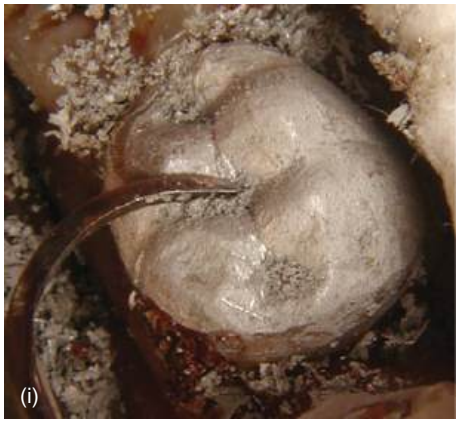


Figure 8.9
(i) Refine the final occlusal morphology and anatomy using carvers (Ward's or Half-Hollenback), sickle scaler, excavator, and/or Mitchell's trimmer, and check occlusion using articulating paper.

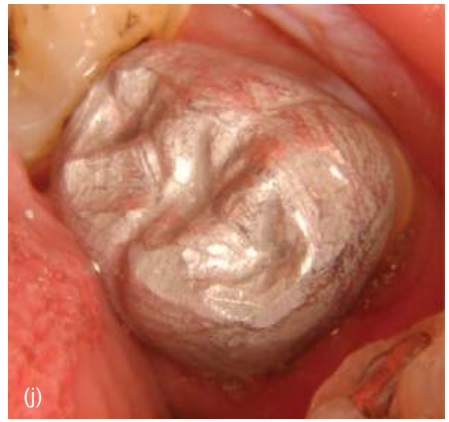


Figure 8.9
(j) After checking the occlusion and margins, the final restoration is burnished and a smooth surface achieved by gently rubbing the surface with a damp cotton wool pledget. (All images courtesy of Dr G. Palmer.)

8.10 'Nayyar core' restoration

Table 8.10 Outline of 'Nayyar core' restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/isolation** (Chapter 5)	**Caries removal/ cavity preparation** (Chapter 5)	**Cavity modification** (Chapters 5 and 6)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
'Nayyar core' restoration Amalgam GIC Composite	Posterior root-filled teeth with short, curved root canals Enough cervical coronal dentine must remain to support the final restoration Technique provides retention and strength for the coronal restoration	Rubber dam isolation if possible Cotton wool rolls/ aspiration	Use already created endodontic access cavity and natural shape of the pulp chamber Remove coronal 2–4 mm of root-canal filling (e.g. gutta-percha) from the obturated canals, using Gates-Glidden burs	Remove any sharp internal line angles with CS rose-head bur Wash and dry final cavity (10 s) Place suitable matrix as required to replace any missing coronal walls of cavity	*Bonded amalgam*: auto-/dual-cure resin cement *GIC:* 10% polyacrylic acid conditioner, 15 s (wash and dry, 10 s) *Composite*: DBA, types 1–4 (see Table 8.12) Original Nayyar core described using dental amalgam, but adhesive materials may be used with their own bonding procedures	Triturate, plug, and condense amalgam into the openings created in the coronal aspect of the root canals using narrow Smith plugger/periodontal probe Pack the remaining cavity, slightly overfill and carve back using appropriate carvers For composites/GICs, ensure the root canal spaces are filled without voids, using a dark shade. Place fine capsule tips into canal orifice and back fill (GIC). Light cure composite increments (470 nm, 20 s)	Remove rubber dam/ matrix band Check interproximal margins with floss and remove overhangs/finish surfaces with amalgam/ composite finishing strips Check occlusion with articulating paper and adjust high spots with carvers/amalgam burs (amalgam), fine diamond burs/discs (GIC/ composite) Burnish final amalgam/ finish with diamond-impregnated polishing discs (GIC/composite)

8.11 Direct fibre-post/resin composite core restoration

Table 8.11 Outline of a direct fibre-post/composite core restoration

Operative procedure	**Indication** (Chapters 2 and 3)	**Preoperative procedures/ isolation** (Chapter 5)	**Caries removal/cavity preparation** (Chapter 5)	**Cavity modification** (Chapters 5 and 6)	**Bonding steps** (Chapter 7)	**Restorative steps**	**Finishing steps**
Direct fibre-post/ composite core restoration (see Figure 6.3 in Chapter 6)	Broken down anterior (and posterior) root-filled teeth Relatively straight root canal(s) Technique retains large adhesive coronal restoration and reinforces remaining coronal/radicular tooth structure	PA radiograph of root-filled tooth Choose shade of composite Rubber dam isolation is ideal. Otherwise, cotton wool and suction are essential	Remove gutta-percha root filling using Gates-Glidden burs. Calculate radicular post length from original length of post minus the length of post required coronally. Take into account root canal length, ensuring a minimum of 4 mm gutta-percha is retained apically. Use LCPA Gauge diameter of post from PA radiograph. Prepare post hole using supplied post drills (narrow-wide diameter in stages) Wash and dry post hole (paper points). Holding fibre-reinforced post with college tweezers, try post in for size and fit Section post to required length (if necessary) using a carborundum disc, from coronal end	Round off any sharp line angles on the coronal surface Wash post hole with NaOCl and dry with paper points Clean fibre-post with alcohol and air dry for 5 s	Use self-etching dual-cure resin cement (e.g. RelyX™ Unicem (3MESPE)) Activate capsule, automix, attach fine nozzle, and backfill post hole	Insert fibre-post, hold in place, remove excess cement with flat plastic instrument and light cure (470 nm, 40 s) or wait 5 min Apply DBA (types 1–4; see Table 8.2) on to coronal tooth surface and coronal portion of post Place interproximal clear matrix strips to separate adjacent teeth/help to create contact points Incrementally place resin composite, building up dentine/enamel shades if chosen, to provisionalize tooth. Start around post, light cure (470 nm) for 20 s and layer next increment Ensure post is covered with composite	Remove rubber dam/matrix strips Check approximal margins with floss—smooth with composite finishing strips Check occlusion with articulating paper and adjust with fine diamond bur, air turbine/ discs Smooth/polish with diamond-impregnated cups/points/discs

8.12 Types of dental adhesives (dentine bonding agents) – a step-by-step practical guide

Table 8.12 Dentine bonding agents: types, constituents, and practical tips for clinical use.

Type	Smear layer	Etch	Primer	Bond
1 **Three-step** Total etch/ etch and rinse	Remove	37% orthophosphoric acid gel painted on enamel (20 s) and dentine (15 s). Removes smear layer, demineralizes (microporosities in enamel) and exposes dentine collagen fi bres. Wash off (10 s) and air-dry (10 s). May notice frosted enamel.	Contains hydrophilic monomer (e.g. HEMA) acting as bi-functional molecules linking hydrophilic collagen to hydrophobic resin monomers, creating the **hybrid layer**. Exists with a carrier – water. Painted on cavity surface with micro-brush (5 s) and gently air-thinned (2–3 s).	Contains HEMA and other, more hydrophobic monomers. Essentially acts as the unfi lled (or in some cases, lightly fi lled) resin composite. This component is painted onto the primer for 5 s and then light cured (470 nm, 20 s). A shiny cavity surface indicates the presence of the cured DBA, ready for fi rst increment of composite.
2 **Two-step** Total etch/ etch and rinse 'moist bonding'	Remove	37% orthophosphoric acid gel painted on enamel (20 s) and dentine (15 s). Removes smear layer, demineralizes (microporosities in enamel) and exposes dentine collagen fi bres. Wash off (10 s). Air-dry (2–3 s). Blot away surface water droplets (cotton wool pledgets, paper points) – **moist bonding**. If enamel appears frosted then it is likely the dentine surface is too dry and should be rehydrated.	Primer and bond mixed together in same bottle – known as the 'adhesive'. The more hydrophilic monomer (e.g. HEMA) is carried in a volatile solvent (e.g. acetone or alcohol) easing its penetration into the moist dentine collagen fi bres, forming the **hybrid layer**. The combined adhesive is rubbed actively into the cavity surfaces using a micro-brush (5 s). The adhesive is air-dried (5 s) to evaporate the solvent and thin the adhesive layer/prevent pooling within the cavity. Visually check the surface for a 'sheen'; a shiny cavity surface indicates the presence of the adhesive. A dull surface indicates that the adhesive has been blown out of the cavity or has been absorbed into the dentine. If dull, a second layer of adhesive is rubbed in (5 s) and air-dried (5 s) and visually checked. Shiny cavity surfaces can be light cured (470 nm, 20 s) and are then ready for composite.	
3 **Two-step** Weaker self-etching primer	Dissolve	The etch and primer have been combined using acidic monomers, creating a self-etching primer. The acidity of these primers is considerably less than that of acid etch. This acidic primer still contains a solvent requiring evaporation, but no washing is required. The acidic primer is applied with the micro-brush on the cavity walls (5 s) and air-dried (5 s), evaporating the solvent/thinning the pooled primer until no rippling of the applied layer is noticed.		The bond has similar chemistry to unfi lled/lightly fi lled resin composite with some hydrophilic monomers included. This adhesive is applied onto the self-etching primer (5 s), air-thinned for 2–3 s and light cured (470 nm, 20 s). A shiny surface appearance indicates the DBA is in place and ready to accept the fi rst increment of composite.
4 **One-step** Strong self-etching primer	Dissolve	All-in-one systems that present the three stages (etch, prime, and bond) together in one application. Stronger acidic monomers (e.g. glycerophosphoric acid dimethacrylate) etch the tooth surface, dissolving the smear layer, demineralizing and, in dentine, exposing the collagen fi bres to the more hydrophilic monomers to penetrate and form the hybrid layer. DBA premixed and rubbed into the cavity surfaces with a micro-brush (5 s). Air-thinned 2–3 s, visually check for shiny surfaces/no rippling. Light cured (470 nm, 20 s).		

8.13 Checking the final restoration

Once any dental restoration has been placed and finished (of varying size, shape, and material), it is wise for the operator to get into the habit of running through a final clinical checklist before discharging the patient. This involves evaluating the following:

- *occlusion/occlusal morphology/marginal ridge position* (assessed using articulating paper pre- and post-operatively; see Chapter 5)
- *marginal adaptation* (positive or negative ledges, overhangs, excess flash material—assessed using straight probe, Briault probe, or dental floss proximally)
- *tight contact points* (for Class II/III restorations—assessed using dental floss)
- *surface finish* (smooth finish—assessed using a dental probe)
- *aesthetics* (especially in the anterior aesthetic zone, colour/translucency co-assessed by the patient).

8.14 Patient instructions

Depending on the type of restorative material used, the patient should be given appropriate instruction on how to look after/manage the new restoration for the immediate 24- to 48-hour period. In all cases, suitable oral hygiene must be achieved. If local anaesthetic has been used, the patient should be warned to take care when chewing, biting, or consuming hot beverages until the effects have worn off.

- For temporary restorations and amalgam, care must be taken to avoid heavy occlusal loading. Ask the patient to avoid chewing too heavily on the filling for 24 hours.
- For dental resin composites, no further instruction is required. The patient should have been forewarned about the possible shade changes as the adjacent teeth rehydrate over the few hours immediately after the rubber dam has been removed (see Figure 8.7y).

8.15 PubMed keywords

www.ncbi.nlm.nih.gov/pubmed/advanced

QR code image 8.1 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address .

Keywords

dental restoration failure; dental restoration longevity; restoration repair